

5(a)(i)	<i>any one from:</i> water ; mineral ions ; A any named appropriate ion AVP ; e.g. plant hormone	1
5(a)(ii)	lignin ;	1
5(b)(i)	<i>any three from:</i> as temperature increases, rate of water vapour loss from leaf increases ; A transpiration for water vapour loss <i>idea that</i> more heat energy absorbed by leaf ; more (heat) energy to evaporate water molecules / increased temperature for latent heat of vaporisation ; A more energy to break hydrogen bonds higher rate of evaporation / AW, from, surfaces / walls, of spongy mesophyll cells ; water vapour molecules have more kinetic energy ; higher rate of diffusion of water vapour out of leaf / steeper water potential gradient ; more stomata are open at higher temperatures so more water vapour is able to diffuse out of leaf ;	3
5(b)(ii)	<i>any one from:</i> cotton leaves have fewer stomata (per unit area) ; <i>idea that</i> cotton leaves have a thicker (waxy) cuticle ; cotton leaves have trichomes ; A hairs for trichomes cotton leaves are curled ; A rolled / folded cotton leaves have, sunken stomata ; A other examples e.g. stomata in, grooves / crypts / chambers / pits cotton leaves have, multilayered epidermis / hypodermis / thick walled epidermis ; AVP ; e.g. suggestion that cotton leaves only have stomata on lower epidermis smaller stomatal aperture	1
5(c)(i)	<i>any one from:</i> more than one nucleus in each cell / more chromosomes ; larger cells ; more organelles ;	1

Question	Answer	Marks
5(c)(ii)	any one from: nucleus will contain more chromosomes / more than 2 copies of each chromosome ; A more DNA molecules / more chromatin AVP ; e.g. <i>ref. to</i> polyploidy (proportionately) larger nuclei	1

Question	Answer	Marks
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